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CRISPR-Cas System: A New Dawn to Combat Antibiotic Resistance

Muhammad Shahzad Rafiq¹ · Muhammad AbuBakar Shabbir² · Ahmed Raza³ · Shoaib Irshad³ · Andleeb Asghar⁴ · Muhammad Kashif Maan⁵ · Mushtaq Ahmed Gondal⁶ · Haihong Hao¹

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Abstract

Antimicrobial resistance (AMR) can potentially harm global public health. Horizontal gene transfer (HGT), which speeds up the emergence of AMR and increases the burden of drug resistance in mobile genetic elements (MGEs), is the primary method by which AMR genes are transferred across bacterial pathogens. New approaches are urgently needed to halt the spread of bacterial diseases and antibiotic resistance. Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR), an RNA-guided adaptive immune system, protects prokaryotes from foreign DNA like plasmids and phages. This approach may be essential in limiting horizontal gene transfer and halting the spread of antibiotic resistance. The CRISPR-Cas system has been crucial in identifying and understanding resistance mechanisms and developing novel therapeutic approaches. This review article investigates the CRISPR-Cas system's potential as a tool to combat bacterial AMR. Antibiotic-resistant bacteria can be targeted and eliminated by the CRISPR-Cas system. It has been proven to be an efficient method for removing carbapenem-resistant plasmids and regaining antibiotic susceptibility. The CRISPR-Cas system has enormous potential as a weapon against bacterial AMR. It precisely targets and eliminates antibiotic-resistant bacteria, facilitates resistance mechanism identification, and offers new possibilities in diagnostics and therapeutics.

Key Points

Antimicrobial resistance is a growing crisis, projected to cause 10 million annual deaths by 2050.

Diverse delivery methods of CRISPR-Cas systems include plasmid vectors, phage vectors, nanoparticles, and electroporation.

CRISPR-Cas systems hold promise for identifying resistance mechanisms and clinical applications, serving as a potent tool against antimicrobial resistance.

1 Introduction

Antimicrobial resistance (AMR) in bacteria is a growing public health crisis that could lead to 10 million annual deaths by 2050 [1]. The World Health Organization (WHO), along with several other groups and experts, acknowledges that the spread of AMR is a serious problem that requires a global, coordinated action plan. According to a 2022 study in the Lancet journal, bacterial AMR was responsible for approximately 4.95 million deaths in 2019, with around 1.27 million deaths explicitly linked to it [2]. A significant threat to global health is posed by ESKAPE pathogens,

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which include *Enterococcus faecium*, *Klebsiella pneumoniae*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, and *Enterobacter species*. These pathogens have developed antibiotic resistance, resulting in fewer treatment options and an increased disease burden. In 2019, they caused 3.57 million deaths related to AMR. Thus, a coordinated global response is needed for surveillance and combating AMR [3].

Horizontal gene transfer (HGT) is considered one of the significant causes of antibiotic resistance development. The proliferation of Gram-negative bacteria that have developed resistance to various antimicrobials, and, in some cases, to all available treatments, poses a significant threat to public health [4]. The current shortage of new antimicrobials is primarily attributed to economic challenges and disincentives in the pharmaceutical industry. Traditional drug development approaches face financial risks, limited profitability, and lengthy regulatory processes, discouraging investment in novel antibiotics. In preventing the spread of antibiotic resistance and regulating horizontal gene transfer, CRISPR-Cas may have great significance [5]. Drug-resistance mechanisms in pathogens are often linked to genetic changes and horizontal gene transfer (HGT) via mobile genetic elements (MGEs) [6]. Due to the rising prevalence of drug-resistant genes such as *bla*_{NDM-1}, *bla*_{KPC}, *mcr-1*, and *bla*_{VIM}, as highlighted in the study by Wu et al., there is an urgent demand for novel therapeutic approaches to effectively address Gram-negative bacterial infections that have acquired resistance to various pharmaceutical agents [7]. AMR can be overcome by developing new drugs or alternative techniques such as combination therapy, nanotechnology, bacteriophage therapy, or gene editing [8]. Bacteria developed CRISPR-Cas and restriction-modification systems through evolutionary selection, impacting phage infection and antibiotic resistance gene spread [9]. According to He et al., the CRISPR-Cas9 technology effectively removes carbapenem-resistant plasmids and returns organisms to their previous level of antibiotic susceptibility [10]. The CRISPR-Cas9 system was introduced through transformation experiments, and sensitivity to antimicrobials was restored by removing the extended-spectrum beta-lactamases (ESBLs) resistance plasmid. Evidence shows that CRISPR-Cas9 can effectively quell the horizontally transferable colistin resistance genes on *mcr-1* plasmids [11]. The CRISPR-Cas9 technology has been shown to concurrently target two drug-resistant genes (*bla*_{NDM-1} and *mcr-1*), allowing for the simultaneous elimination of these genes and restoring the effectiveness of the antibiotics colistin and carbapenem [12]. In the quest to combat bacterial infections and address the global challenge of antimicrobial resistance, the integration of CRISPR/Cas-based detection technologies has emerged as a promising avenue for precise and rapid identification [5].

The therapeutic strategy of CRISPR-Cas faces key challenges, including off-target effects, ethical concerns related to genetic manipulation, and potential unintended consequences. Precision, safety, and ethical considerations are critical areas of focus for the advancement and widespread application of CRISPR-Cas in therapeutic interventions. This article aimed to explore the potential of the CRISPR-Cas system as a tool for combating bacterial AMR. The article discusses the CRISPR-Cas mechanism, various key types, the pivotal role played by different Cas proteins to overcome this global challenge, delivery methods, ongoing CRISPR-based antibacterial clinical trials, and potential future applications of the CRISPR-Cas system to combat AMR. It offers a comprehensive overview of this cutting-edge field in the fight against antibiotic resistance.

2 Mechanism of the CRISPR-Cas System and its Classification

The CRISPR-Cas system consists of repeats, spacers, and CRISPR-associated genes. Repeats are short DNA sequences repeated multiple times within the CRISPR array. Spacers are unique DNA sequences within the CRISPR array between the repeats [13]. Each spacer represents a small fragment of foreign DNA that bacteria encountered in the past, such as phages or plasmids. The organism's immune system acquires these spacers as a memory of previous encounters with foreign genetic material. Cas genes, often called CRISPR-associated genes, are found in conjunction with the CRISPR sequence in bacteria. These genes encode proteins that are essential for the CRISPR-Cas system functioning. These proteins can recognize and cleave the DNA [14]. Figure 1 shows the structure of CRISPR array.

CRISPR/Cas systems encompass a tripartite framework with three distinct phases, constituting the bacterial defensive mechanism [15]. The initial phase of adaptation involves the acquisition of spacer sequences. The second expression stage involves the crRNA and Cas protein production. The last interference step involves the cleavage of nucleic acids directed by the crRNA [16]. The mature CRISPR RNA (crRNA) associates Cas protein to create a nucleic acid and protein complex. This complex can identify nucleic acid sequences that are complementary to the crRNA and exhibit endonuclease activity, resulting in the degradation of the nucleic acid close to the identified site [17]. The inherent specificity of the CRISPR system facilitates its application in targeting and eliminating drug resistance genes, thereby effectively regulating horizontal gene transfer and mitigating the dissemination of antibiotic resistance [18]. Although CRISPR-Cas9 and CRISPR Cas12a (Cpf1) are well known for their use in genome editing, they have been used in the study of antibiotic resistance. To make bacteria more vulnerable to antibiotics, scientists have utilized

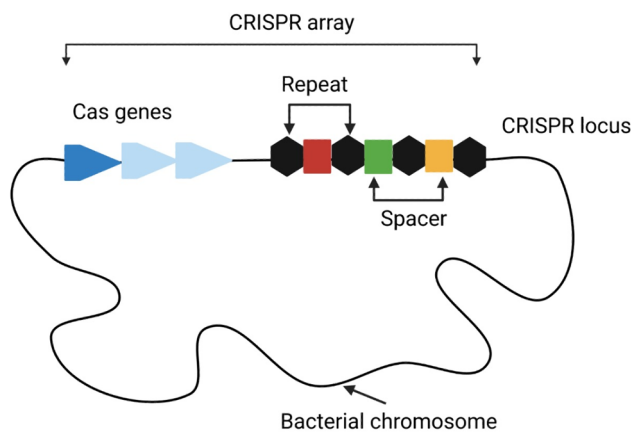


Fig. 1 Structure of the CRISPR array. The black diamonds in the representation shows repeats, while the colourful rectangles are used to indicate spacers. Cas genes, repeats and spacers collectively form a CRISPR array. Created with BioRender.com. *Cas genes* CRISPR-associated genes, *CRISPR* Clustered Regularly Interspaced Short Palindromic Repeats

CRISPR-Cas9 to damage genes responsible for antibiotic resistance specifically.

The CRISPR-Cas system exhibits diverse forms, categorized by the abundance and diversity of Cas proteins. These systems broadly fall into two classes: Class 1 (including types I, III, and IV) characterized by multiple Cas proteins, and Class 2 (encompassing types II, V, and VI) featuring a single gene that typically encodes a large protein binding to CRISPR RNA [19]. There are approximately six types of CRISPR-Cas systems, comprising 36 subtypes. All type I CRISPR-Cas systems feature the Cas3 protein, crucial for eliminating unnatural genetic sequences. In Type III systems, Cas10 targets both RNA and DNA, while Type IV, represented by Csf1, lacks Cas1 and Cas2, typically found on plasmids. Type II systems, characterized by Cas9, process CRISPR RNA and cut DNA. The indispensable Cas proteins include Cas12 (formerly Cpf1) in Type V and Cas13a in Type VI, the latter exclusively targeting foreign RNA sequences [20]. In Fig. 2, an overview is presented depicting the diverse categories of CRISPR-Cas systems, their respective subtypes, pivotal proteins associated with various CRISPR-Cas variants, and the specific genetic material they target, along with corresponding model organisms.

3 Relationship of CRISPR CAS and Antibiotic Resistance

The research indicated a negative correlation between the type II CRISPR-Cas system and antibiotic resistance in *E. faecalis* [21]. Typically, only the orphan CRISPR2 is present in *E. faecalis* isolates. The CRISPR1-Cas and

CRISPR3 systems are generally inactive in *E. faecalis* strains resistant to multiple drugs [22]. Antibiotic resistance and the absence of virulence factors are both linked to the existence of the CRISPR-Cas system. The association between the presence of CRISPR3 in *E. faecalis* and a lack of antibiotic resistance raises intriguing questions about whether this phenomenon is specific to certain antibiotics or if it is a broader consequence of the bacterium's general plasmid recognition and defense mechanisms. Multiple studies have linked the presence of CRISPR3 in *E. faecalis* to a lack of antibiotic resistance in this bacterium [23]. There may be species-specific variations in CRISPR activity against antibiotic resistance genes due to differences in CRISPR1 loci observed among *E. faecalis* species. Conjugated administration of CRISPR-Cas9 has decreased sequence-specific antibiotic resistance in the *Enterococcal* population [24]. It was demonstrated by Price et al. that genomic resistance against MGE at orphan CRISPR2 loci required the presence of CRISPR1-Cas produced from *E. faecalis* [25]. It has been shown that CRISPR-Cas can prevent the acquisition of some corresponding pathogenic factors in some isolates of *E. faecalis*, as presented in a study by Gholizadeh et al., who discovered that the CRISPR-Cas9 technology effectively decreases antibiotic resistance to *E. faecalis* in laboratory and clinical settings by targeting the tetracycline resistance gene (*tetM*) and the erythromycin resistance gene (*ermB*) [26]. According to these studies, CRISPR-Cas systems may be able to control the development of drug resistance in Gram-positive bacteria.

4 Role of Cas Proteins in Eliminating Antibiotic Resistance Genes

4.1 Cas3

Cas3 is a critical component of many bacteria's Type I CRISPR-Cas system. It is a nuclease-helicase enzyme with endonuclease and ATPase activity. During the CRISPR interference stage, Cas3 is primarily responsible for destroying target DNA [27]. By employing the CRISPR-Cas mechanism, specifically Cas3, it has become feasible to identify and eliminate plasmids that harbor antibiotic resistance selectively. Consequently, this approach effectively mitigates the dissemination of antibiotic resistance among bacterial communities. The utilization of this technique holds promise in paving the way for innovative strategies in combating antibiotic resistance [28].

Gomaa et al. employed the type I-E CRISPR-Cas system found in *Escherichia coli*, which encompasses six Cas genes distributed across two operons, namely casABCDE

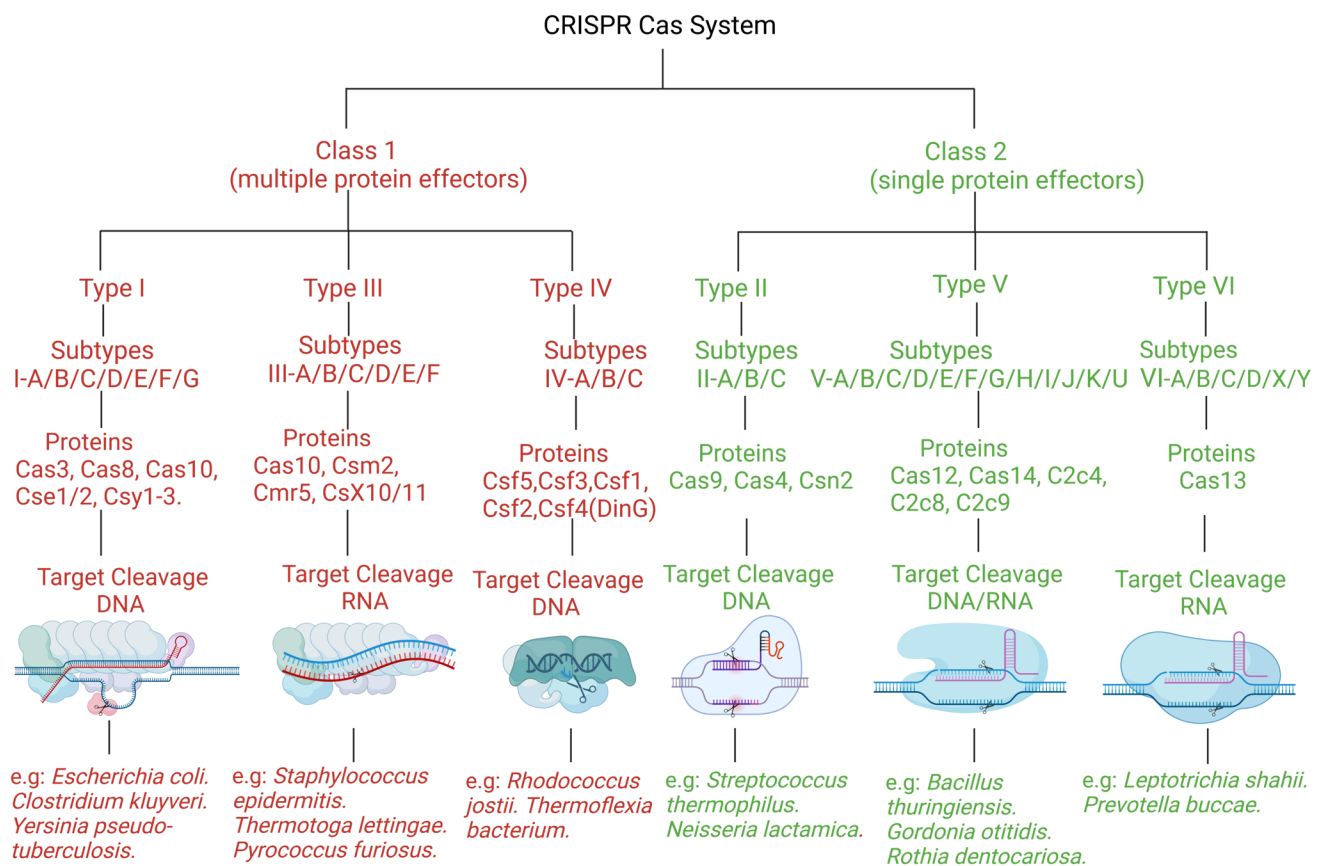


Fig. 2 Various categories of CRISPR-Cas systems, their subtypes, significant proteins to target DNA/RNA with model organisms. Created with BioRender. *Cas* protein CRISPR-associated protein, *Cse* CRISPR-associated endonuclease, *Csf* CRISPR-associated factor,

Csm CRISPR-associated module, *Csn* Nmeni specific-subtype cas protien, *Cmr* CRISPR cas module for RNA-mediated targeting, *DNA* deoxyribonucleic acid, *RNA* ribonucleic acid

and Cas3. This system was utilized to selectively target crucial genomic loci in diverse sequences. The researchers utilized the highly homologous strain *E. coli* B, which is known for concurrent resistance to antimicrobials under specific conditions, while *E. coli* K-12 is a common drug-sensitive laboratory strain. By specifically targeting *fucP* and *ogr*, the researchers successfully eliminated the *K-12* and *B* strains from plasmids containing the CRISPR system. Targeting *groL* resulted in the complete eradication of both strains, indicating the effective removal of drug resistance rather than the killing the bacteria [29]. Furthermore, Selle et al. harnessed the type I-B CRISPR-Cas system from *C. difficile* as a tool to combat antibiotic resistance. The CRISPR-Cas3 system demonstrated significant efficacy in both preventing and treating *C. difficile* infections. This was achieved by utilizing phages loaded with crRNAs, which selectively targeted bacterial chromosomes, ultimately leading to bacterial eradication [30]. In another study, the CRISPR cascade genes (*cse1*, *cse2*, *cas7*, *cas5*, and *cas6e*) and Cas3 of the *E. coli* type I-E CRISPR system were amplified using PCR (polymerase chain reaction). These genes produce enough

proteins to destroy DNA molecules that code for specific protospacers. The supplied CRISPR-Cas system eliminates plasmids that provide antibiotic resistance and genetically altered lytic phages [31].

4.2 Cas 9

Cas9 protein can successfully cause double-strand breaks in plasmid DNA when guided by a specific guide RNA (gRNA) corresponding to a target region on the plasmid. This activates the cellular repair mechanism, potentially leading to plasmid degradation or loss. Several studies in diverse bacterial species have shown successful targeting and eradication of antibiotic-resistant plasmids utilizing the Cas9 mechanism [32].

Despite promising achievements, leveraging Cas9 for plasmid targeting and cure remains difficult. Among these are the possibility of plasmid evolution and the emergence of Cas9-resistant plasmids [33]. Potential plasmid evolution and the emergence of Cas9-resistant variants present a major obstacle to using CRISPR-Cas9 for plasmid targeting

[34]. This risk results from the potential for minor nucleotide alterations in the targeted plasmid sequences, which could cause mutations that obstruct Cas9 recognition and cleavage. Research has in fact brought attention to the risk of developing resistance to CRISPR-based treatment methods [35, 36]. The target organism, the particular CRISPR system used, and the makeup of the targeted genetic elements are some of the variables that affect the frequency of resistance. Certain bacteria may naturally have adaptive mechanisms that make them more prone to developing resistance. One crucial factor to take into account is the necessary sequence identity for CRISPR-Cas9 recognition and activity [37]. Because the guide RNA (gRNA) and target DNA sequences are meant to be complementary, even a slight mismatch between them can cause the CRISPR-Cas9 system to function less effectively or fail to cleave the desired location. Plasmids with mutations that evade Cas9 targeting may survive and proliferate as a result of this phenomenon.

Strategies for overcoming these obstacles are being actively researched, such as using several gRNAs to target key plasmid areas or generating novel Cas9 variations. Furthermore, the creation of Cas9 and gRNA delivery systems that successfully target bacterial cells is an important topic of ongoing research.

There have been several attempts to eradicate resistance genes in diverse species. The study demonstrated the significance of the *bla*_{OXA-55} like gene in conferring carbapenem resistance and reversing antibiotic resistance in *Shewanella algae*. This was achieved through the introduction of a single plasmid carrying both CRISPR/Cas9 and *recE/recT* recombinase. These findings suggest a potential methodology for the process of confirming the accuracy and correctness of the assigned functions to various genes within the genome of *Shewanella algae* [7]. The study demonstrated the potential use of plasmids utilizing the *IncP RK2* conjugative system as carriers for a *TevSpCas9* dual nuclease within a co-culture system of *E. coli* and *Salmonella enterica*. It is noteworthy that, in contrast to a trans system where the conjugation and CRISPR components are separate, a cis-acting plasmid containing both elements demonstrated a higher frequency of *E. coli* conjugation to *S. enterica* [38]. Another study assessed the efficacy of phage treatment in experimental infections caused by S242, an *E. coli* strain responsible for deadly newborn meningitis and globally disseminated, which carries the extended-spectrum *beta-lactamase CTX-M-15*. From ambient water, EC200(PP), a lytic phage that is aggressive against S242, was identified [39].

The use of Cas9 protein in the targeting and curing of antibiotic-resistant plasmids is a potential approach to addressing antibiotic resistance. Cas9's capacity to precisely edit plasmid DNA and its variable specificity makes it an

effective tool for selectively removing antibiotic resistance genes [40].

4.3 Cas13

According to a 2016 study by Abudayyeh et al., CRISPR-Cas13a displays promiscuous ssRNA cleavage capabilities and limits the development of host bacteria by degrading their RNAs [20]. This has proven to be a phage defense mechanism [41]. When phages infect bacteria, CRISPR-Cas13a detects the phage genome's transcript, which causes non-specific destruction of the bacterial transcripts and halts the development of the bacterial cell to stop the infection from spreading [42]. Four distinct CRISPR-Cas13a subtypes have recently been discovered by researchers using 11 strains of six different *Leptotrichia* species. One of these, CRISPR-Cas13a from *Leptotrichia shahii* (*LshCas13a*), significantly inhibited bacterial growth [20].

Kiga et al. also showed the targeted lethal effect of CRISPR-Cas against *E. coli* in 2020. The sequence-specific bacterial medication was based on CRISPR-Cas13a, which made a difference. Additionally, the Cas9-based bactericide is only applied to bacteria having the target gene on their chromosome. The Cas13a-based agent, on the other hand, kills bacteria with the target gene on both their chromosome and a plasmid. In the study conducted by Kiga et al., the CRISPR-Cas13a system was employed in an *E. coli* strain carrying the carbapenem-resistant gene *bla*_{IMP-1}. This system demonstrated a higher bactericidal efficacy compared to CRISPR-Cas9. Furthermore, the CRISPR-Cas13a mechanism does not break bacterial DNA directly. Instead, it explicitly targets bacterial mRNA, which has lower mutation activity. This makes CRISPR-Cas13a highly promising for development and use as an antibacterial agent [43]. The lower mutation activity of mRNA suggests that, as a target for CRISPR-Cas13a, it may offer a more stable and reliable approach for precision in bacterial RNA interference without introducing as many variations or errors in the genetic code. This characteristic enhances the specificity and effectiveness of the CRISPR-Cas13a mechanism in targeting and regulating bacterial gene expression. Figure 3 depicts the CRISPR-Cas gene-editing process in target bacteria. It illustrates the steps involved, including the identification of the target gene (*ARGs*). The gRNA guides the Cas protein to precisely cleave the target gene (*ARGs*), inducing mutations that make the bacteria more susceptible to antibiotics. Table 1 summarizes the role of different Cas proteins in eliminating antibiotic resistance in different bacteria with different delivery methods.

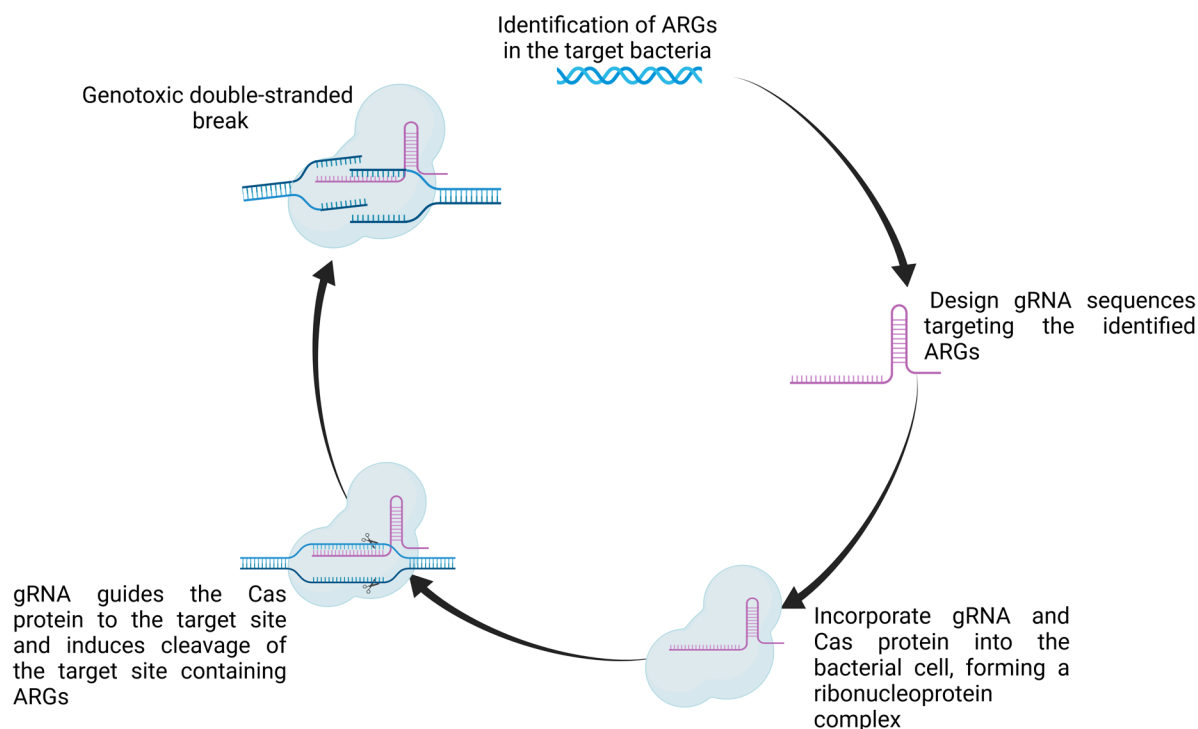


Fig. 3 Illustration of CRISPR-Cas gene editing process in bacteria: The gRNA guides the Cas protein to precisely cleave the target gene (ARGs), inducing mutations that make the bacteria more susceptible

to antibiotics. Created with Biorender. ARGs antibiotic-resistance genes, gRNA guide ribonucleic acid, Cas protein CRISPR-associated protein

5 Strategies to Deliver CRISPR-Cas Systems

The study by Wan et al. highlighted that the CRISPR-Cas system holds promise for regulating and thwarting the dissemination of drug-resistant genes, thereby addressing microbial resistance concerns [44]. The CRISPR-Cas system is employed by guide RNA to specifically recognize drug-resistant genes and guide their cleavage in targeted regions. Consequently, this leads to the elimination of bacteria or the restoration of their susceptibility to antibiotics. Delivering CRISPR-Cas systems into bacteria faces inherent challenges, including the cell wall barrier, the need for precise and efficient delivery methods, and the risk of off-target effects. Overcoming these challenges is essential for successful genome editing in bacterial cells. Several delivery strategies of CRISPR-Cas delivery have been developed, each with its own set of benefits, drawbacks, and considerations.

5.1 Delivery of CRISPR-Cas by Plasmid Vector

By incorporating the CRISPR-Cas9 sequences or a few sequences of its components into a vector plasmid, the CRISPR-Cas9 system is used to reduce the prevalence of antibiotic-resistant bacteria. Conjugation is a crucial method for transferring bacterial genes, promoting the dissemination of resistance genes. Pyne et al. reported that antibiotic

resistance in *Enterococcal* populations could be reduced by sequence-specific delivery plasmids expressing the bacteriophage CRISPR-Cas9 [45]. It restores the sensitivity of the resistance bacteria to carbapenems and shows beneficial therapeutic effects during clinical isolation. According to Kim et al., when the conjugation plasmid is effectively transferred to recipient bacteria, it has the potential to facilitate the dissemination of the CRISPR-Cas9 system to other bacterial strains. Targeted antimicrobial plasmids (TAPs) can sensitize recipient cells expressing *pOXA*, halting drug resistance. This approach effectively delivers the CRISPR-Cas9 system to *E. coli* and *Enterobacteriaceae* [12]. The utilization of plasmid vectors for delivering CRISPR-Cas9 has gained significant popularity due to its exceptional efficiency. Plasmid vectors can efficiently enter target bacteria and deliver CRISPR-Cas9 components into the target bacteria, including the Cas9 nuclease and the guide RNA, which enables efficient genome editing and manipulation [46]. Circular DNA molecules called plasmid vectors are used to introduce CRISPR-Cas9 components into target bacteria in order to edit their genomes. The Cas9 gene and guide RNA (gRNA) sequence are carried by these vectors. When the plasmid is inserted into bacteria, it replicates on its own, directs the Cas9 nuclease to particular genomic regions, and causes double-strand breaks. The target DNA is subsequently altered by the cell's repair machinery, allowing for

Table 1 Antibiotic resistance - targeting genes, eliminating resistance, and enhancing bacterial susceptibility

Cas protein	Target bacteria	Delivery method	Target gene	Function of gene	Ref
Cas3 (Type I)	<i>C. difficile</i>	Phage	Bacterial chromosome	–	[81]
	<i>E. coli</i>	Phage	<i>bla</i> _{NDM-1} CTX-M-15	β -lactam resistance β -lactam resistance	[31]
	<i>E. coli</i>	Plasmid	<i>fucP</i> gene, <i>ogr</i> gene, <i>groL</i> gene	L-Fucose-H+ Membrane Transport Protein, Protranscriptional regulator OgrK, Chaperone protein GroEL	[29]
Cas9 (Type II)	<i>S. algae</i>	Plasmid	<i>sul2</i> , <i>bla</i> _{OXA-55} like, <i>NmcR</i> -like	Sulfonamide resistance β -lactam resistance β -lactam resistance	[7]
	<i>E. coli</i> , <i>Salmonella</i>	Plasmid	65 genes*	38 Essential genes, 23 nonessential genes and 4 genes with unresolved phenotype	[38]
	<i>E. faecalis</i>	Plasmid	<i>ErmB</i>	Macrolide resistance	[89]
	<i>K. pneumoniae</i>	Electroporation	<i>tetA</i> , <i>ramR</i> , <i>mgrB</i>	Tetracycline resistance Tetracycline resistance Polymyxin resistance	[90]
	<i>S. aureus</i>	Electroporation	<i>tarO</i> , <i>tarH</i> , <i>tarG</i>	Lysostaphin resistance	[91]
	<i>Enterobacteriaceae</i> species	Electroporation	<i>bla</i> _{KPC}	β -lactam resistance	[33]
	<i>S. aureus</i>	Mobile genomic island	<i>agrA</i>	Cylo-lipopeptide resistance	[92]
	<i>S. aureus</i>	Phage	<i>nuc</i>	Nuclease	[93]
	<i>S. aureus</i>	Nanoparticle	<i>mecA</i>	β -lactam resistance	[94]
	MRSA	Phage	<i>mecA</i>	β -lactam resistance	[95]
	<i>E. coli</i>	Plasmid (heat shock)	<i>fosA3</i>	Fosfomycin resistance	[96]
	<i>E. coli</i>	Plasmid	<i>mcr-1</i>	Polymyxin resistance	[97]
	<i>E. coli</i>	Plasmid	<i>bla</i> _{OXA-48}	β -lactam resistance	[98]
	<i>E. coli</i>	Plasmid	<i>bla</i> _{TEM-1}	β -lactam resistance	[99]
	<i>E. coli</i>	Plasmid	<i>bla</i> _{NDM-5}	β -lactam resistance	[100]
	<i>E. coli</i>	Phage	<i>bla</i> _{SHV-18} <i>bla</i> _{NDM-1}	β -lactam resistance, Metallo- β -lactam resistance	[101]
		Phage	<i>bla</i> _{NDM-1}	Metallo- β -lactam resistance	[102]
Cas13a (Type- VI)	<i>E. coli</i>	Phage	<i>bla</i> _{IMP-1}	β -lactam resistance	[43]

AgrA Accessory gene regulator A, *BlaIMP-1* Beta-lactamase IMP-1, *BlaKPC* *Klebsiella pneumoniae* carbapenemase, *BlaNDM-1* New Delhi metallo, *BlaNDM-5* New Delhi metallo-beta-lactamase-5, *BlaOXA-48* oxacillinase-48, *BlaOXA-55* oxacillinase-55, *BlaSHV-18* sulfhydryl variable beta-lactamase 18, *BlaTEM-1* Temoniera beta-lactamase-1, *CTX-M-15* Cefotaxime-Munich 15, *Cas* CRISPR-associated, *Cas3* CRISPR-associated protein 3, *Cas9* CRISPR-associated protein 9, *Cas13a* CRISPR-associated protein 13a, *CRISPR* Clustered Regularly Interspaced Short Palindromic Repeats, *E. coli* *Escherichia coli*, *E. faecalis* *Enterococcus faecalis*, *K. pneumoniae* *Klebsiella pneumoniae*, *L-Fucose-H+* L-fucose-proton symporter, *MRSA* methicillin-resistant *Staphylococcus aureus*, *NDM-1* New Delhi metallo-beta-lactamase-1, *ND* not determined, *Nuc* nuclease, *PHA* polyhydroxyalkanoates, *S. algae* *Shewanella algae*, *S. aureus* *Staphylococcus aureus*, *Sul2* sulfonamide resistance gene 2, *TarO*, *TarH*, *TarG* teichoic acid ribitol-2'-O-methyltransferase O, teichoic acid ribitol-2'-O-methyltransferase H, teichoic acid ribitol-2'-O-methyltransferase G, *TetA* tetracycline resistance protein A, *ErmB* erythromycin resistance methylase B

*These 65 genes are grouped around the *S. enterica* chromosome, are unresolved phenotype genes, non-essential genes, and essential genes. The 65 genes are significant because CRISPR-based antimicrobial therapy may be able to target them. Using particular single-guide RNAs (sgRNAs) delivered by a conjugative plasmid containing the CRISPR-Cas9 system, these genes were targeted

precise genetic alterations like gene insertion or knockout. In molecular biology, plasmid vectors are crucial instruments for effective and regulated genome editing. Plasmid vectors can hold long DNA sequences, including multiple Cas9 variants, guide RNAs, and other regulatory elements. Because of this adaptability, researchers can customize and optimize the CRISPR-Cas9 system for specific applications, tailoring it to their experimental requirements [47]. Another benefit

of using plasmid vectors for CRISPR-Cas9 delivery is their simplicity. Standard molecular biology techniques can efficiently design and manipulate these vectors in the laboratory. They can be built, modified, and propagated without complicated procedures, making them available to researchers [48]. Cpf1 generates staggered double-strand breaks in the target DNA, creating a different cleavage pattern compared to the blunt ends produced by Cas9 [49]. These attributes make

Cpf1 a valuable addition to the CRISPR toolbox, providing researchers with a diverse range of options for genome editing and manipulation. When delivered via a plasmid vector, Cpf1 holds immense potential for precise genetic engineering in various organisms [50]. The delivery of CRISPR-Cas systems via plasmid vectors is hampered by inherent drawbacks. These include low stability, which can lead to reduced expression of CRISPR components, and low conjugation efficiency, potentially limiting the number of cells that successfully receive the genome-editing machinery [46]. Conjugation potential can vary between and within bacterial species. Some bacterial species exhibit high conjugation efficiency, meaning that they are more likely to successfully transfer genetic material to other cells. This property is influenced by the presence of specific conjugative plasmids and the compatibility between the donor and recipient cells. The potential of bacteria to transfer genetic material between species may have consequences for the safety and effectiveness of CRISPR-based therapies when used in conjunction with in vivo treatment approaches. For instance, there may be unintended or off-target effects if the therapeutic plasmids encoding CRISPR components are transferred to unintended bacterial species. To overcome this limitation, researchers may consider repeated plasmid delivery or investigate alternative delivery methods to achieve long-term effects.

5.2 CRISPR-Cas Delivery by Phage Vector

Phage vectors exhibit superior efficiency in infecting host bacteria compared to plasmid vectors. They possess the capacity to transport larger DNA fragments, thereby enabling the incorporation of multiple proteins into the CRISPR-Cas9 system. The encapsulation of nucleic acid within phage proteins confers resistance to enzymatic degradation [51]. Pyne et al. used phage as a delivery technique due to its advantages in naturally targeting microorganisms. Utilizing bacteriophages as vectors, guide RNA (sgRNA) was designed targeting conserved sequences within *beta-lactamase* genes. This approach led to the elimination of over 200 harmful bacteria carrying altered forms of *beta-lactamase* and effectively restored their sensitivity to beta-lactam antibiotics [45]. Researchers employed lysogenic phages in the creation of a distinctive type I CRISPR-Cas3 mechanism. This mechanism effectively eradicated drug-resistant plasmids and reinstated the host bacteria's sensitivity to a range of antibiotics. These findings indicate the potential of a new, phage-based CRISPR-Cas9 system for delivering and targeting bacterial drug resistance plasmids [52]. Cpf1 generates staggered double-strand breaks, which can lead to more precise and predictable editing outcomes. Its smaller size compared to Cas9 facilitates packaging into phage vectors, making it a practical choice for efficient delivery. The combination of Cpf1 and phage-based delivery systems

holds great potential for advancing the field of genome editing with enhanced specificity and efficiency [53]. Utilizing phages for CRISPR-Cas9 delivery offers distinct advantages, including its unique mechanism and notably high efficiency in introducing the system. CRISPR-Cas9 delivery via phage vector is highly efficient for several reasons. Phages have evolved to infect specific bacterial hosts efficiently, ensuring the effective delivery of CRISPR-Cas9 components. This targeted delivery method enables precise and efficient bacterial genome editing [54]. Speed is another significant advantage of phage-based delivery. Phages have a short replication cycle; they can quickly deliver and edit the bacterial genome. Therefore, it can be instrumental in time-sensitive experiments or applications that require quick results [55]. The delivery of CRISPR-Cas systems via phage delivery is constrained by its limited host range, restricting its applicability to specific bacterial species. The surface limitations of phages may hinder their ability to efficiently deliver CRISPR components to target cells, potentially reducing the effectiveness of the genome-editing process [56]. The application of CRISPR-Cas systems is restricted to particular bacterial species due to the limited host range that impedes phage delivery. Phage surface restrictions might make it more difficult for CRISPR components to be delivered effectively, which could lessen the efficacy of genome editing. A significant drawback to this strategy is surface binding common resistance to phages.

5.3 Nanoparticle-Based Delivery of CRISPR-Cas

Nanoparticles can also deliver CRISPR-Cas, as they can easily cross biological membranes due to their tiny size. When the CRISPR-Cas9 system is encased in nanomaterials, it is protected from degradation and may be distributed efficiently and selectively [57]. Various nanomaterials, including polymeric nanocrystals and nanoparticles, cationic lipid nanoparticles, nanocages, and more, have all been studied for their potential as antibiotic transporters. To improve the efficiency of editing bacterial genomes targeting *mecA*, Rodrigues et al. combined Cas effector proteins that can be successfully transported into resistant *S. aureus* [58].

One drawback is that nanoparticles may cause immunological reactions, which might have undesirable side effects or reduce the effectiveness of drug delivery [59]. On the other hand, improvements in nanoparticle design have attempted to reduce immunogenicity. Particle size, surface charge, and stability must all be carefully optimized for creating effective nanoparticles for CRISPR-Cas9 delivery. In one study, researchers loaded Cas13 protein and gRNA into lipid nanoparticles and demonstrated efficient delivery and knockdown of specific RNA targets in cells. The nanoparticle-mediated delivery enhanced the stability and intracellular availability of the CRISPR-Cas13 components [60].

Delivery of CRISPR-Cas via nanoparticle-based carriers presents significant other drawbacks, including inherent low stability of the nanoparticles in physiological environments, leading to potential degradation before reaching target cells. These carriers often exhibit suboptimal efficiency in delivering CRISPR components, and may induce undesirable levels of cell toxicity, underscoring the need for further refinement in nanoparticle design for enhanced therapeutic efficacy [61]. Nanoparticles can penetrate the outer membrane of Gram-negative bacteria, which is a significant barrier due to its complex structure. The small size of nanoparticles allows them to penetrate these barriers, making them versatile carriers for delivering CRISPR-Cas components to a wide range of bacterial species, regardless of their Gram classification. For the best CRISPR-Cas delivery, possible immunological responses and issues with the stability and effectiveness of nanoparticles must be addressed. In terms of bacterial uptake, nanoparticles can deliver their cargo for genome editing to bacteria by means of endocytosis or direct penetration.

5.4 Delivery of CRISPR-Cas Through Electroporation

An electroporation-based CRISPR-Cas9 methodology was utilized in a study to propose the first genome-editing technique for *R. eutropha*. This significantly increased the efficiency and decreased the time needed to regulate this facultative chemolithoautotrophic bacterium. This novel method allows for a more in-depth study of *R. eutropha* and applications for PHA (polyhydroxyalkanoates) production and CO₂ conversion [62].

Wu et al. achieved the reinstatement of the Lst-induced clearance impact on *S. aureus* by employing electroporation to introduce a plasmid harboring CRISPR-dCas9 into the *S. aureus* strain ATCC 6538 [63]. Nevertheless, challenges persist in terms of cytotoxicity and cellular harm, limiting its suitability for in vivo applications. In addressing *S. aureus* infections, Ram et al. pioneered the creation of antibacterial community (ABD) systems based on engineered staphylococcal pathogenicity islands (*SaPIs*), which are more suitable for in vitro studies [64]. Using *SaPIs* as the vector, ABDs enact CRISPR-Cas9, whose spacers target the *agr* gene [65]. While in vivo research has shown the direct elimination of infected *S. aureus* by ABDs, sustaining system stability over the long term remains a challenge [66]. Proteins other than Cas9 have also emerged as promising candidates for electroporation-based delivery of CRISPR-Cas systems. Cpf1 (also known as Cas12), an RNA-guided endonuclease with characteristics distinct from Cas9, is one notable example. Cpf1 is well known for its ability to recognize various protospacer adjacent motif (PAM) sequences, thereby broadening the targetable genomic regions [67]. Cpf1 causes staggered double-strand breaks, which may allow for more efficient

homology-directed repair (HDR) processes. This property improves the precision of genome-editing applications [68]. The delivery of CRISPR-Cas by electroporation can lead to reduced cell viability due to the inherent stress and damage induced by the electrical pulses. This method is often limited in its transformation efficiency, making it less suitable for large-scale applications where high-throughput delivery of genetic material is required [69]. Electroporation-based CRISPR-Cas delivery has the potential to treat targeted infections by introducing gene-editing tools into bacteria. However, challenges such as cytotoxicity and long-term stability must be addressed for effective therapeutic use in vivo. More research and optimization are required to improve this approach's viability in treating infections. Figure 4 shows the delivery of CRISPR-Cas through nanoparticle, plasmid, phage, and electroporation-based delivery methods.

6 CRISPR/Cas System as a Tool for Resistance Mechanism Identification

The CRISPR-Cas system can be utilized to identify resistance mechanisms in bacteria. In the context of bacterial resistance, researchers can use CRISPR-Cas systems to better understand and uncover the mechanisms through which bacteria develop antibiotic resistance [70]. This method enables scientists to discover previously identified genes or pathways that lead to resistance, offering essential insights for creating novel therapeutic techniques. Resistance-related genetic elements, such as mobile genetic elements (plasmids) or resistance genes, may be precisely edited or added using CRISPR-Cas systems. This allows researchers to look into how these components spread among bacterial populations and contribute to resistance spread [15]. Using CRISPR-Cas technology to study bacterial resistance mechanisms is a powerful tool for comprehending the genetic basis of antibiotic resistance and developing effective anti-resistance strategies.

The FLASH (Finding Low Abundance Sequences by Hybridization) toolbox represents a significant advancement in CRISPR-Cas AMR detection [71]. FLASH identifies low abundance AMR genes in complex microbial communities by combining CRISPR-based target enrichment with next-generation sequencing [72]. The practical use of CRISPR-Cas systems, particularly through the FLASH toolbox, represents a significant advance in the fight against antimicrobial resistance. This technology offers a powerful and adaptable method for detecting AMR genes with unprecedented accuracy and speed [73]. FLASH capitalizes on the high efficiency, specificity, and adaptability of Cas9 to selectively amplify a predetermined collection of sequences [74]. The FLASH-NGS method significantly enhances enrichment, reaching up to five orders of magnitude, and enables the

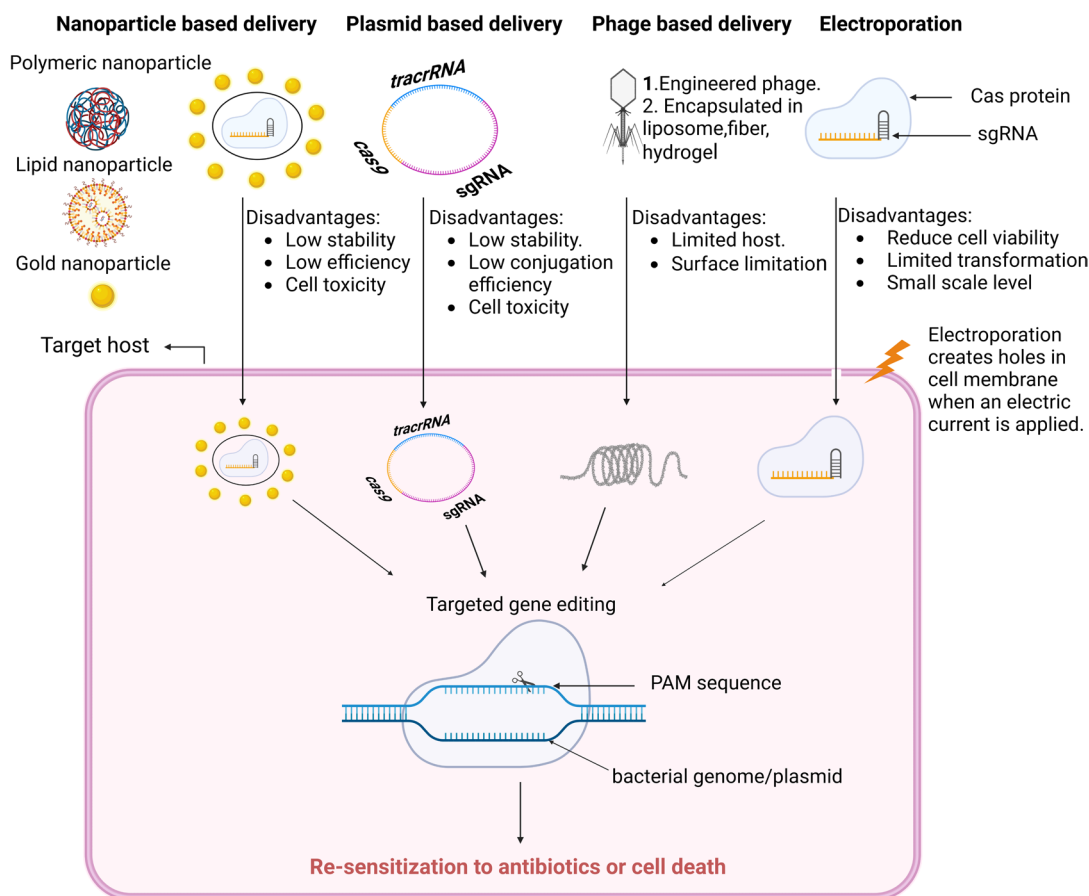


Fig. 4 Delivery of CRISPR-Cas through nanoparticle, plasmid, phage and electroporation-based delivery. In nanoparticle-based CRISPR gene editing, CRISPR Cas9 editors are initially encapsulated within nanoparticles. Three distinct types of nanoparticles, including gold nanoparticles, polymeric nanoparticles, and lipid nanoparticles, are employed to deliver CRISPR Cas9. Nanoparticles assist Cas9 by reducing recognition, effectively protecting it from degradation, facilitating release, enabling assembly in the target bacteria, and carrying out the gene editing process. Plasmid-based delivery: Through the utilization of conjugative plasmid systems such as targeted anti-bacterial plasmids (TAPs), it becomes possible to deliver both the CRISPR-Cas system and single guide RNA (sgRNA) sequences. This can be done with the aim of either directly eliminating the bacteria or preventing the expression of antibacterial resistance genes. Phage

delivery is common for CRISPR-Cas, but faces two issues: limited host range and surface pathogen targeting. To address this, mutate the phage's fibrin for wider host range and use encapsulation in liposomes, fibers, or hydrogels to boost effectiveness. The delivery of CRISPR-Cas through electroporation involves the application of brief electrical pulses to create temporary pores in the cell membrane. This process allows for the efficient uptake of CRISPR-Cas molecules and facilitates precise gene editing within the cells. The delivery methods offer distinct advantages in terms of efficiency, specificity, and safety for various applications. Created with BioRender. *Cas9* CRISPR-associated protein 9, *Cas protein* CRISPR-associated protein, *sgRNA* single-guide RNA, *tracrRNA* trans-activating Cluster Regularly Interspaced Short Palindromic Repeat Ribonucleic Acid, *PAM sequence* Protospacer Adjacent Motif sequence

detection of genes at sub-attomolar levels while maintaining little background noise. Several factors could impact the performance of FLASH-NGS. These factors encompass the complexity of the guide RNA set, input DNA quantity, and Cas9 protein [75]. While challenges remain, ongoing research and development efforts promise to further refine and expand the use of CRISPR-Cas systems in combating this urgent global health threat. As the field develops, it is expected that CRISPR-based AMR detection will play a critical role in preserving the efficacy of antimicrobial agents for future generations.

7 CRISPR/Cas-Based Detection of Bacteria

Polymerase chain reaction (PCR) stands as a time-honored technique for nucleic acid amplification, and its application continues to grow in the realm of CRISPR-based diagnostic technologies [76]. The accurate detection of infections relies heavily on advanced technologies capable of identifying the nucleic acids of infectious agents. One highly accurate method is the real-time polymerase chain reaction (RT-qPCR). However, its practical application on-demand might be hindered due to the considerable expenses of acquiring reagents and specialized equipment. As a result,

there is a substantial need for low-cost methods that may be easily used at the point of care (POC). Isothermal nucleic acid amplification is a promising solution in this context, despite having slightly less selectivity than PCR. Its advantage lies in avoiding the need for thermal cycling, which helps reduce costs and makes it more feasible for POC applications [77]. Consequently, it is of utmost importance to devise approaches that merge the accuracy of PCR with the simplicity and cost-effectiveness of isothermal amplification. CRISPR-Cas is an excellent candidate for this purpose, a remarkably selective and reprogrammable technology that can potentially create POC diagnostic tests reliant on nucleic acids.

In 2017, Kellner and a team of researchers devised the SHERLOCK diagnostic method, which effectively detects nucleic acids by harnessing the cleavage ability of Cas13 on ssRNA molecules [78]. SHERLOCK employs reverse transcription RPA (RT-RPA) to amplify nucleic acids, specifically DNA or RNA. Once amplified, the resulting amplicon is equipped with a T7 promoter, enabling the generation of RNA molecules from it. Subsequently, the Cas13a enzyme from *Leptotrichia wadeii* (LwaCas13a) forms a complex with a crRNA that possesses a complementary sequence to the target. Upon binding to the target molecule, the Cas13a-crRNA complex initiates cleavage of both the target molecule and the single-stranded RNA reporter molecules. These RNA reporter molecules are linked to fluorophores and quenchers. The cleavage of the fluorophore and quencher leads to the production of a fluorescence signal. This highly sensitive method allows for the detection of viral, bacterial, and human nucleic acids with attomolar sensitivity [78]. SHERLOCKv2 enabled simultaneous measurement and detection of nucleic acids from various sources by detecting cleaved reporter molecules using antibody-conjugated gold nanoparticles on a paper strip. Notably, SHERLOCK and HUDSON's integration to navigate epidemics (SHINE) (heating unextracted diagnostic samples to obliterate nucleases) has successfully sensed SARS-CoV-2 in a one-step reaction with 90% sensitivity and 100% specificity compared to RT-qPCR, which also detects antibiotic resistance elements [79].

Numerous Cas12-based platforms have been developed to detect microorganisms. Among these, a highly sensitive and specific method has been established by combining CRISPR-Cas12a with a label-free impedance assay. This integrated approach enables the identification of *E. coli* and *S. aureus*, with an impressive limit of detection (LOD) of 3 nM for *S. aureus* [80]. Moreover, a novel approach integrating Recombinase Polymerase Amplification (RPA) and CRISPR-Cas12 has successfully detected carbapenem-resistance genes, including *bla_{KPC}*, *bla_{NDM}*, and *bla_{OXA}*, within a remarkably short duration of less than 1 h. This technique demonstrates comparable sensitivity and accuracy

to RT-qPCR, ensuring reliable and rapid identification of these genes [81]. The correlation between the fluorescence test and affordable portable strips has been established successfully. A novel method called HOLMES (an abbreviation for 1-h, low-cost, multifunctional, highly efficient system) has been developed for nucleic acid detection, which utilizes LbCas12a and PCR for pre-amplification. In its second iteration, HOLMES incorporates LAMP and Cas12b from *Allicyclobacillus acidoterrestris* (AacCas12b) to detect targets in a single container, achieving an approximate detection limit of 10 aM [82].

Ge et al. created a Cas14a1-based nucleic acid detection platform (CMP) for rapidly identifying microorganisms in substances such as milk. The method involves utilizing the collateral activity of the Cas14a1-sgRNA complex to cleave non-specific ssDNA attached to fluorescent and quenching agents when the target ssDNA is detected. This approach successfully quantified *S. typhi* and *S. pyogenes* in samples with higher sensitivity compared to RT-qPCR [83]. In a novel approach, Wei et al. introduced an innovative platform for highly accurate detection of live *S. aureus* using an aptamer-based Cas14a1 biosensor (ACasB), completely bypassing the need for nucleic acid extraction or amplification. The method involved the initial hybridization of a DNA blocker with the *S. aureus* specific aptamer. The blocker was released after the introduction of live *S. aureus* due to its binding with the bacterial aptamer. This event led to a noticeable shift in fluorescence intensity, triggering the activation of the Cas14a1 protein through its interaction with the sgRNA. ACasB demonstrated rapid and reliable differentiation of 400 CFU/mL live *S. aureus* cells, exhibiting excellent specificity and sensitivity. Remarkably, the Cas14a1-aptamer biosensor achieved 100% accuracy in detecting *S. aureus* in challenging samples, comparable to the widely used RT-qPCR method [84]. Table 2 shows a comparative overview of detection platforms and their features.

8 Clinical Application of the CRISPR System

CRISPR-based diagnostic techniques have been shown to offer a significant amount of potential for diagnosing infectious diseases. The exact diagnosis of infectious bacteria is significant for treatment, control of AMR, and survival of patients. This system is a highly selective, reprogrammable technology that could be utilized to construct nucleic acid-based diagnostic assays. Novel diagnostic platforms using CRISPR-Cas technology have been created to detect pathogens, employing various types of nucleic acids as targets. Following this, the suitable CRISPR-Cas system (such as Cas9, Cas12, Cas13, and Cas14) effectively identifies and cuts the designated target. The outcomes can be visualized through fluorescence, lateral flow tests,

Table 2 CRISPR detection: a comparative overview of detection platforms and their features

Detection platform	Cas protein	Nuclease domain	Amplification	sensitivity	Guide RNA	Cleavage activity	Target	Target region
HOLMES	Cas12a	RuvC and nuclease lobe (NUC) domains	PCR/RT-PCR	aM	crRNA	Yes	dsDNA/ssDNA	PAM sequence (CTA, TTN, TTTN)
DETECTR	Cas12a	RuvC and nuclease lobe (NUC) domains	RPA	aM	crRNA	Yes	dsDNA/ssDNA	PAM sequence (CTA, TTN, TTTN)
HOLMESv2	Cas12b	RuvC and nuclease lobe (NUC) domains	LAMP	aM	sgRNA	Yes	dsDNA/ssDNA	PAM sequence (CTA, TTN, TTTN)
SHERLOCK	Cas13a	2HEPN	RPA	aM	crRNA	Yes	ssRNA	3' PFS
SHERLOCKv2	Cas13b	2HEPN	RPA	zM (10–21)	crRNA	Yes	ssRNA	Degraded RNA is dependent on double side protospacer flanking site (5' end and PAM sequence (NAN/NNA) at the 3' end.
DETECTR	Cas14a	RuvC	RPA	aM	sgRNA	Yes	ssDNA	Cleave without PAM uses only T-rich sequences

Cas12a CRISPR-associated protein 12a, *Cas12b* CRISPR-associated protein 12b, *Cas13a* CRISPR-associated protein 13a, *Cas13b* CRISPR-associated protein 13b, *Cas14a* CRISPR-associated protein 14a, *DETECTR* DNA endonuclease-targeted CRISPR Trans Reporter, HOLMES 1-h low-cost multipurpose highly efficient system, *HOLMESv2* HOLMES version 2, LAMP Loop-mediated Isothermal Amplification, *PAM* Protospacer Adjacent Motif, *PCR* polymerase chain reaction, *RNA* ribonucleic acid, *RPA* recombinase polymerase amplification, *SHERLOCK* Specific High-sensitivity Enzymatic Reporter unlocking, *sgRNA* single guide RNA, *ssDNA* single-stranded DNA, *ssRNA* single-stranded RNA

electrochemical methods, or colorimetric techniques [85]. Multiple research investigations have been conducted on CRISPR-based diagnostics, exploring the role of Cas proteins such as Cas9, Cas12, Cas13, and Cas14 [87].

The CAS-EXPAR method utilizes the CRISPR-Cas9 system to identify DNA methylation and total RNA in *Listeria monocytogenes* utilizing an isothermal exponential amplification reaction [86]. Scientists have achieved a significant breakthrough by developing an exact and accurate method to detect *E. coli* and *S. aureus* infections utilizing CRISPR-Cas12a technology [80]. An efficient detection method utilizing a Cas12a/gRNA system successfully identified both *M. tuberculosis* and a diverse range of six key species belonging to the non-tuberculous mycobacteria (NTM) group [87]. Table 3 explains the applications of the CRISPR system in pathogen detection. The use of CRISPR-Cas antimicrobials is still in its infancy, and there are still many challenges to overcome [88].

Table 4 explains the limitations of the use of CRISPR-Cas systems in the fight against drug-resistant bacterial strains as well as advantages in their use over standard methods.

Table 5 explains different CRISPR-based antibacterials listed in clinicaltrials.gov.

9 Conclusion

The worldwide spread of AMR bacteria poses a significant global health issue. To counteract AMR, many strategies have been devised. One possible method is the CRISPR-Cas system, which provides a formidable tool in the fight against drug-resistant illnesses. The CRISPR-Cas9 system enables the targeting and elimination of antibiotic-resistant bacteria both within and outside of cells. This cutting-edge innovation has the potential to strike at the genetic basis of bacteria that are resistant to antibiotics, providing unprecedented precision and adaptability. As progress in this area proceeds, CRISPR-Cas has the potential to play a major role in resolving a major problem in world health. The application of CRISPR/Cas-based detection systems represents a transformative approach in bacterial diagnostics, offering unparalleled specificity and efficiency that holds immense potential in shaping the future of infectious disease detection

Table 3 Applications of the CRISPR system in pathogen detection

Cas protein	Platform name	Pathogen diagnosed	Pre-amplification method	Signal output	Sensitivity	Time (min)	References
Cas9	CAS-EXPAR	<i>L. monocytogenes</i> mRNA	EXPAR	Fluorescence	82 × aM	60	[86]
Cas9 nickase	Cas9nAR	<i>S. typhimurium</i> , <i>E. coli</i> , <i>M. smegmatis</i>	Strand-displacing DNA polymerase	Fluorescence	17 aM	60	[103]
dCas9	CRISPR-mediated DNA-FISH	MRSA	No	Fluorescence	10 CFU/mL	30	[104]
Cas9	CASLFA	<i>L. monocytogenes</i>	PCR	Lateral flow	150 copies	60	[105]
Cas12a	Cas12a-UPTLFA	<i>Yersinia pestis</i>	RPA	UPT-LFA	3 aM	<1h	[106]
Cas12a	CRISPR-MTB	<i>M. tuberculosis</i> H37Ra strain	RPA	Fluorescence	50 CFU/mL	90	[107]
Cas12a	RAA-Cas12a	<i>S. aureus</i>	RAA	Fluorescence	10 copies/μL	1 h	[108]
Cas12a	CRISPR-PCR	<i>N. cyriacigeorgica</i>	PCR	Fluorescence	10–3 ng	120	[109]
LbCas12a	OCTOPUS	<i>E. coli</i> O157:H7, <i>S. aureus</i>	RPA	Fluorescence	1 CFU/mL	50	[110]
Cas12a	CARID	<i>V. cholerae</i>	Recombinase aided	Fluorescence / lateral flow	20 copies/reaction	–	[111]
Cas12a	RAA-CRISPR/Cas12a	<i>V. vulnificus</i>	RAA	Fluorescence	2 copies/reaction	40	[112]
Cas12a	CRISPR-HP	<i>H. pylori</i>	RPA	Lateral flow	5 copies/μL	60	[113]
Cas12a	ICB-LAMP-CRISPR/Cas12a	<i>C. jejuni</i>	LAMP	Fluorescence	8 CFU/mL	70	[114]
Cas12a	DropCRISPR	<i>S. typhimurium</i>	LAMP	Fluorescence	102 CFU/mL	NR	[115]
LwCas13a	SHERLOCK	<i>Y. pestis</i>	RPA	Fluorescence	420 copies/mL	150	[116]
Cas13	SHERLOCKv2	ZIKV, DENV, <i>P. aeruginosa</i> , <i>S. aureus</i>	RPA	Fluorescence or lateral flow	8.0 zM	120–240	[117]
LwCas13a	CCB-Detection	<i>S. aureus</i>	PCR & reverse transcription	Fluorescence	1 CFU/mL	240	[118]
Cas13a	CRISPR-GBS	<i>S. agalactiae</i>	RPA	Fluorescence	50 CFU/mL	90	[83]
Cas13a	RapidCasD	<i>P. jirovecii</i>	Transcription-mediated amplification (TMA)	Fluorescence	2 copies/μL	80	[119]
LwCas13a	PCF	<i>Salmonella</i> spp.	PCR and reverse transcription	Fluorescence	10 CFU/mL	120	[120]
Cas14a1	CMP	<i>S. pyogenes</i> , <i>E. typhi</i>	Reverse transcription and APCR	Fluorescence	103 & 104 CFU/mL	–	[83]
Cas14a1	Cas-TPSE	<i>E. coli</i> , <i>E. typhi</i> , <i>P. aeruginosa</i> , <i>S. aureus</i> , <i>S. pyogenes</i> , <i>E. faecalis</i>	PCR	Fluorescence		150	[121]

APCR accurate polymerase chain reaction, CFU colony forming unit, CRISPR Clustered Regularly Interspaced Short Palindromic Repeats, EXPAR Exponential Amplification Reaction, FISH Fluorescence In Situ Hybridization, *H. pylori* *Helicobacter pylori*, LAMP Loop-Mediated Isothermal Amplification, LFA lateral flow assay, *M. smegmatis* *Mycobacterium smegmatis*, *M. tuberculosis* *Mycobacterium tuberculosis*, MRSA methicillin-resistant *Staphylococcus aureus*, *N. cyriacigeorgica* *Nocardia cyriacigeorgica*. NR not reported, *P. aeruginosa* *Pseudomonas aeruginosa*, *P. jirovecii* *Pneumocystis jirovecii*, PCR polymerase chain reaction, RAA recombinase polymerase amplification, RPA recombinase polymerase amplification, *S. agalactiae* *Streptococcus agalactiae*, *S. aureus* *Staphylococcus aureus*, *S. pyogenes* *Streptococcus pyogenes*, *S. typhimurium* *Salmonella typhimurium*, *S. aureus* *Staphylococcus aureus*, SHERLOCK Specific High-sensitivity Enzymatic Reporter UnLOCKing, UPTLFA Up-Converting Phosphor Lateral Flow Assay, *V. cholerae* *Vibrio cholera*, *V. vulnificus* *Vibrio vulnificus*, *Y. pestis* *Yersinia pestis*, ZIKV Zika virus

Table 4 Limitations of the use of CRISPR-Cas systems in the fight against drug-resistant bacterial strains and advantages over standard methods

Aspect	Limitations of CRISPR-Cas systems	Advantages over standard methods
Precision	Potential off-target effects	High precision in targeting specific genes or sequences, minimizing collateral damage
Efficiency	Variable editing efficiency	Enhanced efficiency in modifying bacterial genomes
Adaptability	Limited to target-specific genes	Adaptability to new drug-resistant strains through simple reprogramming
Resistance development	Bacteria may develop resistance to CRISPR-Cas systems	Lower likelihood of bacterial resistance compared to antibiotics
Delivery	Effective delivery mechanisms may be challenging	Versatility in delivery methods, including viral vectors and nanoparticles
Off-target effects	Unintended gene modifications	Improved control and reduced off-target effects with advancements in CRISPR technology
Safety concerns	Potential for unintended consequences in host organisms	Ongoing research to address safety concerns and improve specificity
Treatment time	May take time to design and implement customized CRISPR therapies	Rapid development and deployment of CRISPR-based treatments for specific strains
Cost	Initial development costs can be high	Potential cost savings in the long term, especially for recurrent drug-resistant strains
Regulatory challenges	Regulatory approvals and ethical concerns	Potential for regulatory adaptation to accommodate innovative treatments
Global applicability	May require advanced infrastructure	Scalability and adaptability for use in various settings and regions

CRISPR Clustered regularly interspaced short palindromic repeats, *DNA* deoxyribonucleic acid, *RNA* ribonucleic acid

Table 5 Clinical trials of CRISPR-based antibacterials

No.	Title	NCTC number	Condition	Interventions	Status	References
1	Identification of Host Factors of Norovirus Infections in Mini-Gut Model	NCT03342547	Gastrointestinal infection	Duodenal biopsy/ saliva	Unknown	[122]
2	Species-specific Bacterial Detector for Fast Pathogen Diagnosis of Severe Pneumonia	NCT05143593	Severe pneumonia	Diagnostic Test: SSBD	Not yet recruiting	[123]
3	Evaluation of CRISPR-based Test for the Rapid Identification of TB in Pulmonary Tuberculosis Suspects	NCT04074369	Tuberculosis, pulmonary	Diagnostic Test: CRISPR-based test	Unknown status	[124]
4	Detection of Enterovirus Genotypes by CRISPR Technology	NCT04535648	Enterovirus infections	Other: Non-invasive detection method: CRISPR technology	Not yet recruiting	[125]
5	Effect of PCR-CRISPR/Cas12a on the Early Anti-infective Schemes in Patients With Open Air Pneumonia	NCT04178382	Severe sepsis	Diagnostic Test: PCR-CRISPR/Cas12a detection	Unknown status	[126]
6	Pathogenic Bordetella Rapid Detection	NCT04535505	Pertussis	Diagnostic Test	Withdrawn	[127]

Cas12a CRISPR-associated protein 12a, *CRISPR* Clustered Regularly Interspaced Short Palindromic Repeats, *PCR* polymerase chain reaction, *SSBD* species-specific bacterial detector

and management. The necessity for additional CRISPR-Cas technology development and careful evaluation of ethical and safety considerations are just two examples of the

obstacles that must still be overcome. However, the CRISPR-Cas system shows great promise in the battle against antibiotic resistance, giving us reason to believe that our antibiotic

arsenal will be able to continue to be successful in the face of ever-evolving bacterial threats in the future.

Declarations

Ethics Approval and Consent to participate Not applicable.

Consent for Publication Not applicable.

Availability of Data and Materials Not applicable.

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Code Availability Not applicable.

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